DPUSCNXSGE

This DPU controls the graphic representation of a SCNX database with the visualization SCNXSGE.

Mainly, DPUSCNXSGE is not configured using DPU parameters, but using the configuration section “SCNXSGE Params” instead. That section is described in more detail in the “Reference Database SCNX”.

1. Static parameters:

FOVExtraHorizLeft	Angle (in degrees) to the left, within which the landscape is drawn, in addition to the angle that is required by SGEView’s camera settings. Should be set to 60 on a computer whose SGEView looks to the rear right (Cam-Gamma = 120), otherwise to 0. Default: 0
FOVExtraHorizRight	Angle (in degrees) to the right, within which the landscape is drawn, in addition to the angle that is required by SGEView’s camera settings. Should be set to 60 on a computer whose SGEView looks to the rear left (Cam-Gamma = -120), otherwise to 0. Default: 0

2. Dependencies to other DPUs:

DPUSCNXSGE depends on DPUSGEWorld and DPUSCNX. The lowest Index should be assigned to DPUSCNX, then DPUSGEWorld and finally the highest Index to DPUSCNXSGE.

After DPUSCNXSGE, an instance of DPUSGEView is usually created, which shows a view to the simulated world.

3. Functional principle:

The visualization SCNXSGE mainly displays the following objects:

- ground surface
- road and its superstructures
- landscape objects
- Area2 objects

All of these obejcts are only displayed within the visibility radius of the driver, which is defined by the parameters *SCNX Params: SCN H: MaxRadius[XY]* (in meters). The ground surface's visibility can be reduced further using the parameters *SCNXSGE Params: SCNXSGE Ground: MaxRadius[XY]*. However, both parameters are usually set to the same value.

An individual description of the objects follows, and the applicable restrictions for the scenario designer are listed.

1 - Ground surface:

The ground surface extends – in principle – indefinitely into all directions. It is displayed as a height field, i.e. a square grid of polygons whose corners lie on integral X and Y coordinates, where the Z coordinate corresponds to the landscape's height at this point.

The grid's subdivision is performed dynamically, and it is refined near the driver and on slopes. At the same time, the total number of triangles used for the ground surface is limited to the parameter *SCNXSGE Ground: DesiredTri*.

The landscape height is calculated by the height model implemented in SCNX, which is explained in the "Reference Database SCNX" (chapter 'Landscape generators').

If the part of the landscape which is currently visible contains a lot of hills and mountains of different sizes ("complex landscape"), the number of triangles which are required for displaying the slopes of these hills rises. Thus, fewer triangles remain for the smoothing of the ground surface and for small bumps.

In that case, the sudden appearance of bumps can be noticed by the driver when approaching them. To fix this problem, either the landscape complexity must be reduced or the number of available triangles must be increased. In the latter case, please check for stuttering to make sure the system is not overloaded.

The ground surface is drawn basically everywhere and does not have any holes – so, it is also drawn under the road. This may lead to problems in one special situation: If the road is located within a ditch (i.e. the ground on both sides is higher than the road surface) and the landscape is not subdivided enough, the landscape ("grass") may protrude onto the road. There are three possible solutions to this problem:

- Increase the number of available triangles for the ground, if the system performance allows.
- Decrease the landscape complexity around the problematic area. (When driving within a ditch, the landscape is often barely visible.)
- Lower the landscape around the problematic area, for example by using a negative **Offset** parameter in the course definition, at least on one side of the road (either **LandscapeTypeLeft** or **LandscapeTypeRight**). The landscape can rise at a larger distance to the road again without further troubles, for which the **Gradient** parameter can be used.

2 - Road and its superstructures:

The visualization of rural road segments (i.e. instances of road segment type Course) is prepared extensively when loading the simulation. To do so, the road geometry is sampled along its center and each segment is approximated by a quadrilateral. The subdivision is dynamic: The more the road is bent within an area, the more points are used to create a sufficiently detailed representation. This can be influenced using the parameters *SCNXSGE CourseLaneCell: MaxErrorXY* and *SCNXSGE CourseLaneCell: MaxErrorST*.

Depending on the type of superstructure, it is included within the road surface texture (road markings) or separate strips of quadrilaterals are used (guardrails).

3 - Landscape objects:

The landscape objects along Course road segments are placed as SGE objects on the ground surface. For each kind of object that can be placed using a landscape generator, a global object pool exists, from which objects are taken for all road segments that are partially or fully visible[1]. The object pool's size is defined by the parameters *SCNX Params: SCN LandscapeGeneration: [ObjectGroup XXX:] MaxSupply*. The setting means that at most *MaxSupply* instances of the given object type (e.g. one kind of tree) are displayed at the same time. If more objects of a type are requested, for example by high Density values in the *LandscapeType* definition of a Course, they will not be visible.

The total size of the global object pool heavily influences the visualization's frame rate and therefore needs to be matched to the system's performance. Overall, not more than 2000 landscape objects should be visible at the same time, which corresponds to 100 objects per object type.

For some simple objects (especially trees), SCNXSGE uses a particularly efficient display method, which allows to display five times as many objects without increasing the system load. This allows to set *MaxSupply* to 500 for trees, and thus to create dense forests.

4 - Area2 objects:

The Area2 objects created using SILABAEdit are positioned and rendered as separate SGE objects. These are static objects which are not modified dynamically during the simulation.

[1] Objects placed by the user („Objects = {...}") are processed independent of the global object pool and are **definitely** displayed within the visibility range.